

CSc 453: A Three-Address Intermediate Code Instruction Set for C--

The syntax of a three-address intermediate code is suggested below; the document **Translating 3-Address Code to MIPS Assembly Code** ([link](#)) is based on this instruction set. You are not required to use this particular instruction set for your project, but if you decide to use something different you will have to figure out the translation to MIPS assembly code yourself. (This usually isn't difficult to do, but something to be aware of nonetheless.)

Below, n denotes a (decimal) integer constant, and *label* denotes a label.

Assignment and Copy Statements

x := y op z

The usual binary operations: *op* is one of $+$, $-$, $*$, $/$, and **y** and **z** are constants or variables.

x := -y

Unary minus. **y** is a constant or variable.

x := y

A copy operation. **y** is a constant or variable.

Conditional and Unconditional Jumps

goto label

Unconditional jump

if x relop val goto label

val is an identifier or a decimal integer, *relop* is one of $\leq$, $<$, $\geq$, $>$, $==$, $!=$.

label L

L is a label and must be unique in the program.

Instructions to manage function calls/returns

enter f

f is (a pointer to the symbol table entry of) a function.

leave f

f is (a pointer to the symbol table entry of) a function.

param x

x is an actual parameter.

call p, n

p is a function name, *n* the number of arguments.

return

return to the caller.

set_retval *x*

set the value to be returned by the function to *x*.

get_retval *x*

Copy the return value into *x*